ADVANCE COMPUTER NETWORKS

LAB ASSIGNMENT 5

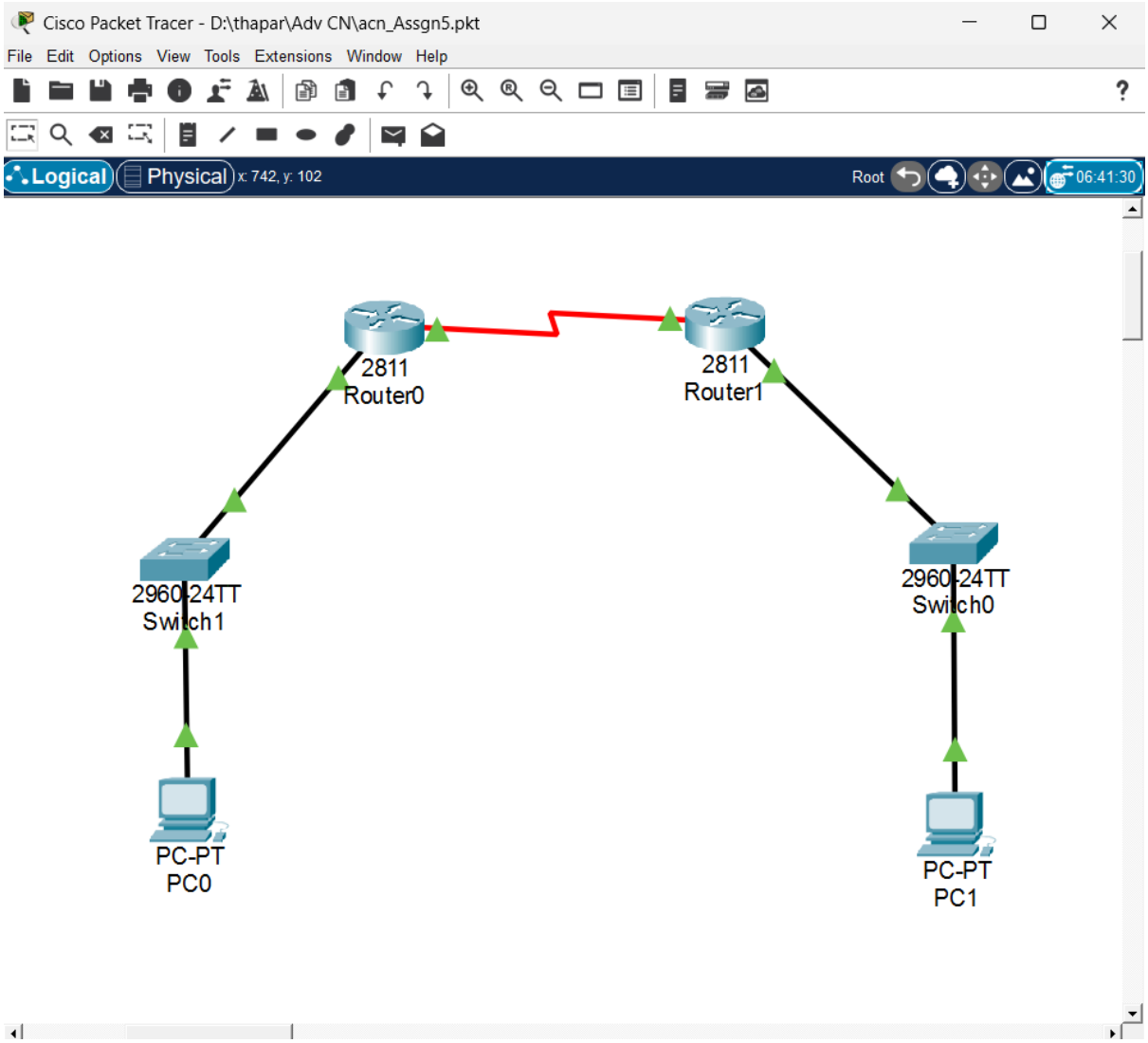
Objective: **Configure Static Route in Cisco packet tracer**

Static route tells the device exactly where to send traffic, no matter what.

Static route is often used when your network has only a few routers or there is only one route from a source to a destination.

Scenario:

Suppose that your company has 2 branches located in Tehran and Shiraz. As the administrator of the network, you are tasked to connect them so that employees in the two LANs can communicate with each other. After careful consideration you decided to connect them via static route.



Router0

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#end'
                  ^
% Invalid input detected at '^' marker.

Router(config)#'
               ^
% Invalid input detected at '^' marker.

Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router0

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Router1

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 172.16.0.1 255.255.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Router0

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.1     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/2/0            unassigned      YES unset  administratively down down
Serial0/2/1            unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.1        YES manual up                    up
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
Router#
```

Router1

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        172.16.0.1      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.2        YES manual up                    up
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
Router#
```

Router0

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 172.16.0.0 255.255.0.0 10.0.0.2
Router(config)#
Router(config)#
```

Router1

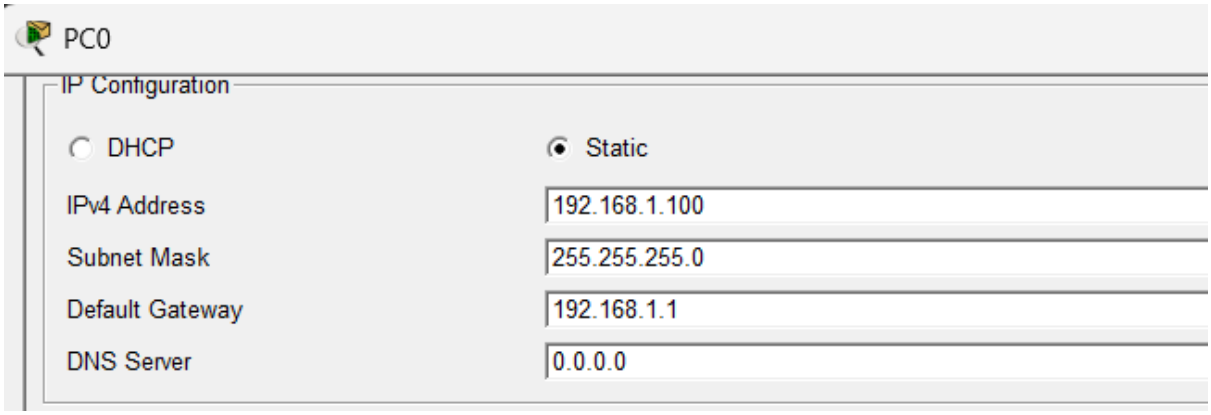
```
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#
```

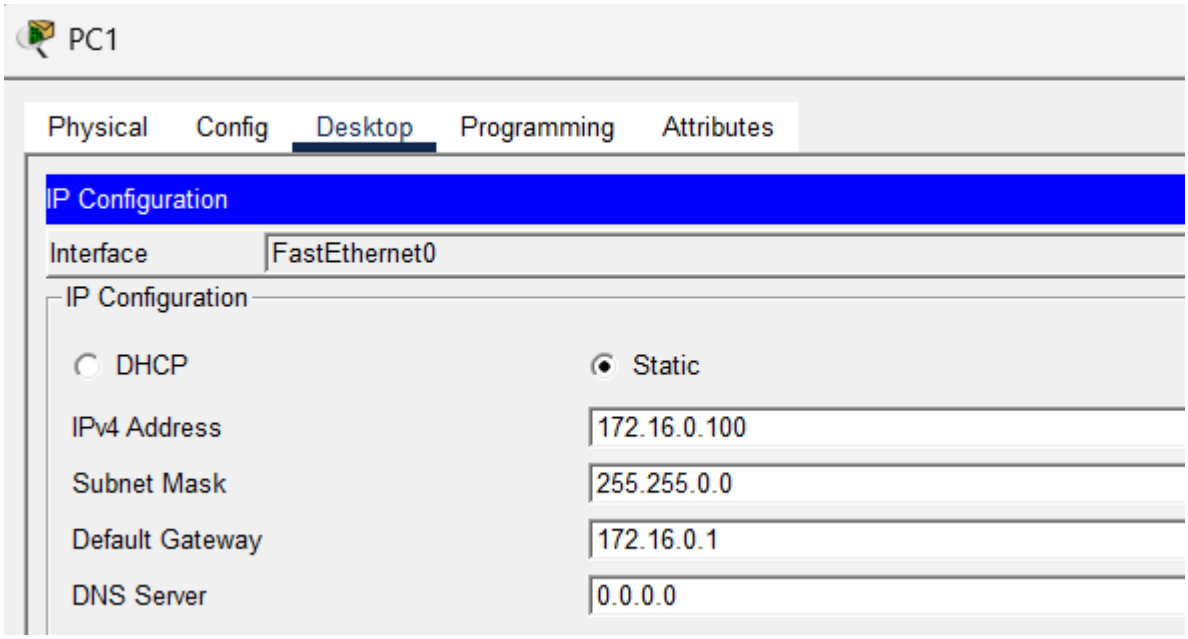
Router0

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
S    172.16.0.0/16 [1/0] via 10.0.0.2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0

Router#
Router#
```

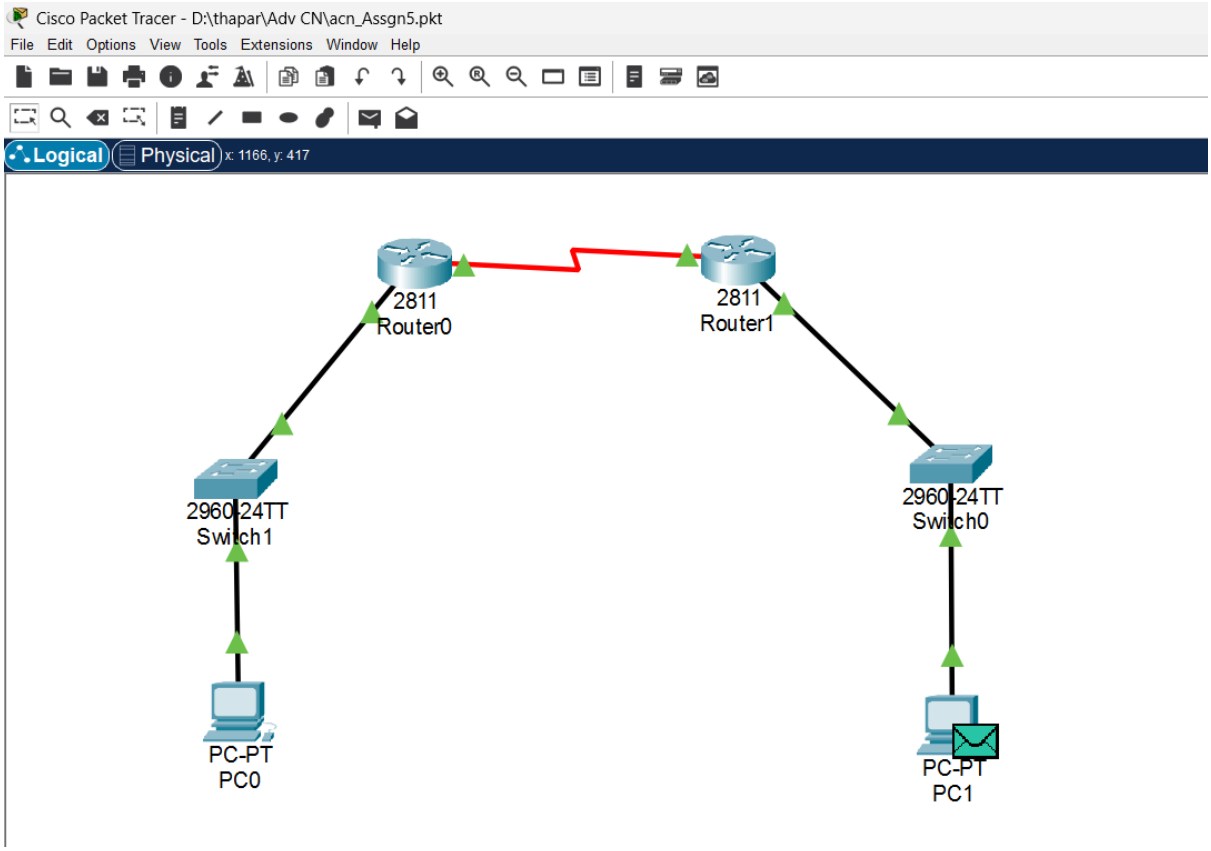
Router1

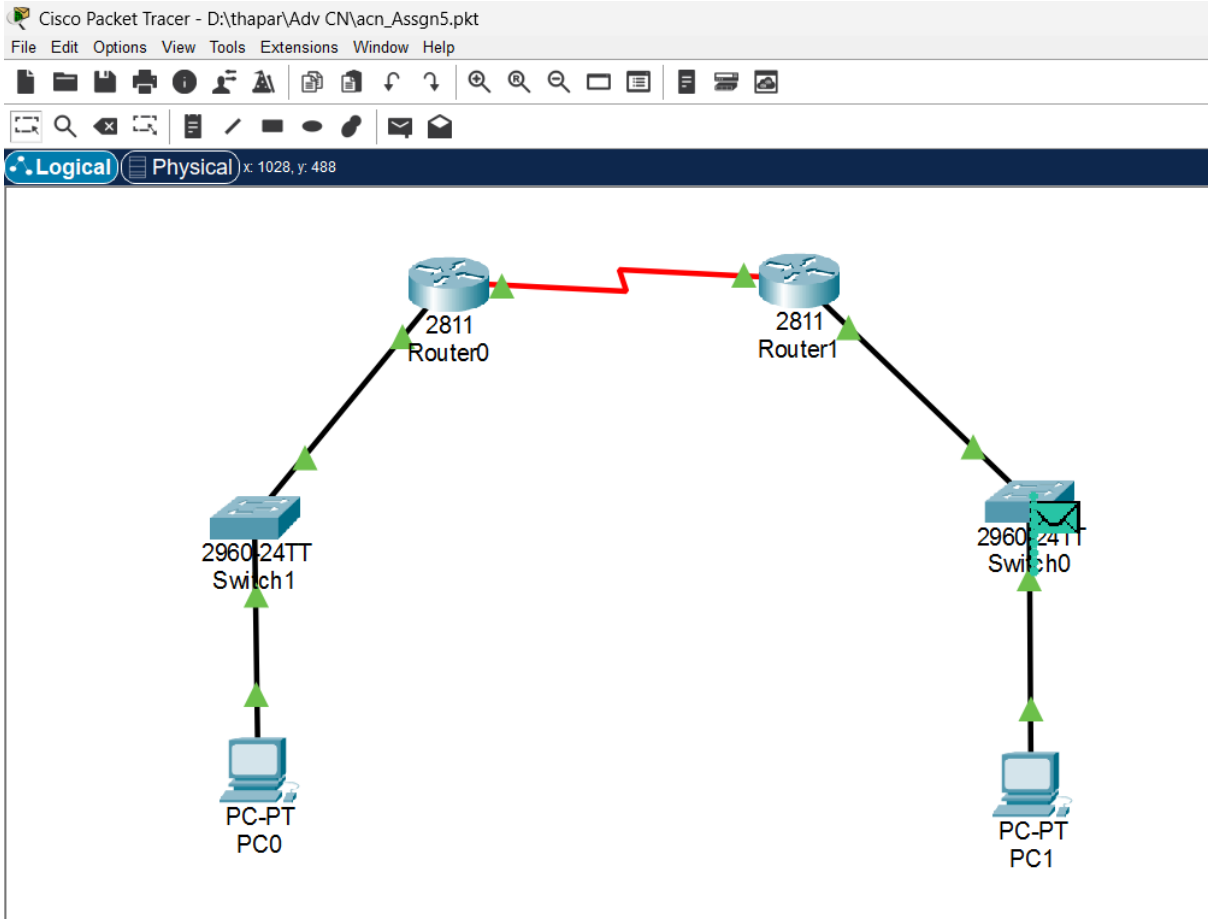
```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.2/32 is directly connected, Serial0/3/0
     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, FastEthernet0/0
L       172.16.0.1/32 is directly connected, FastEthernet0/0
S    192.168.1.0/24 [1/0] via 10.0.0.1

Router#
Router#
```

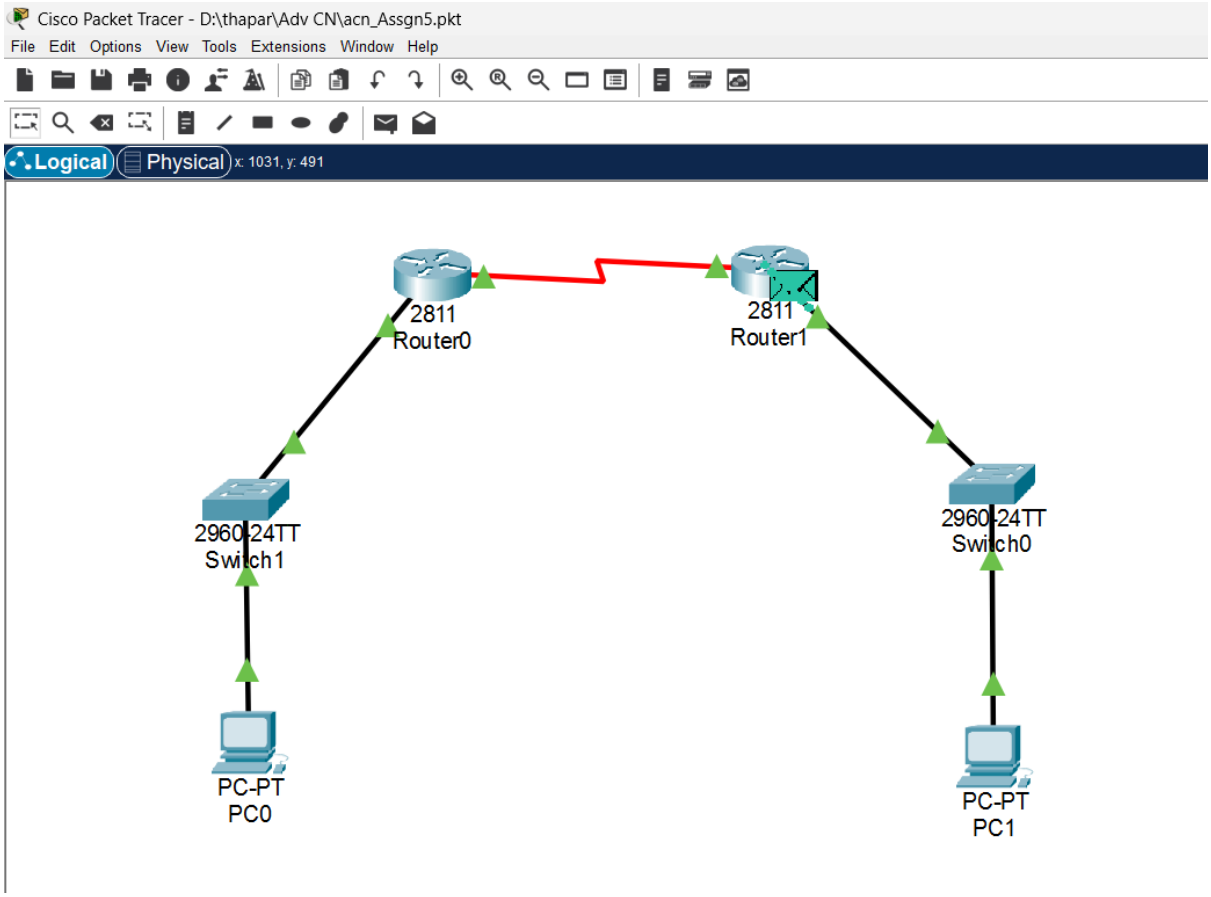


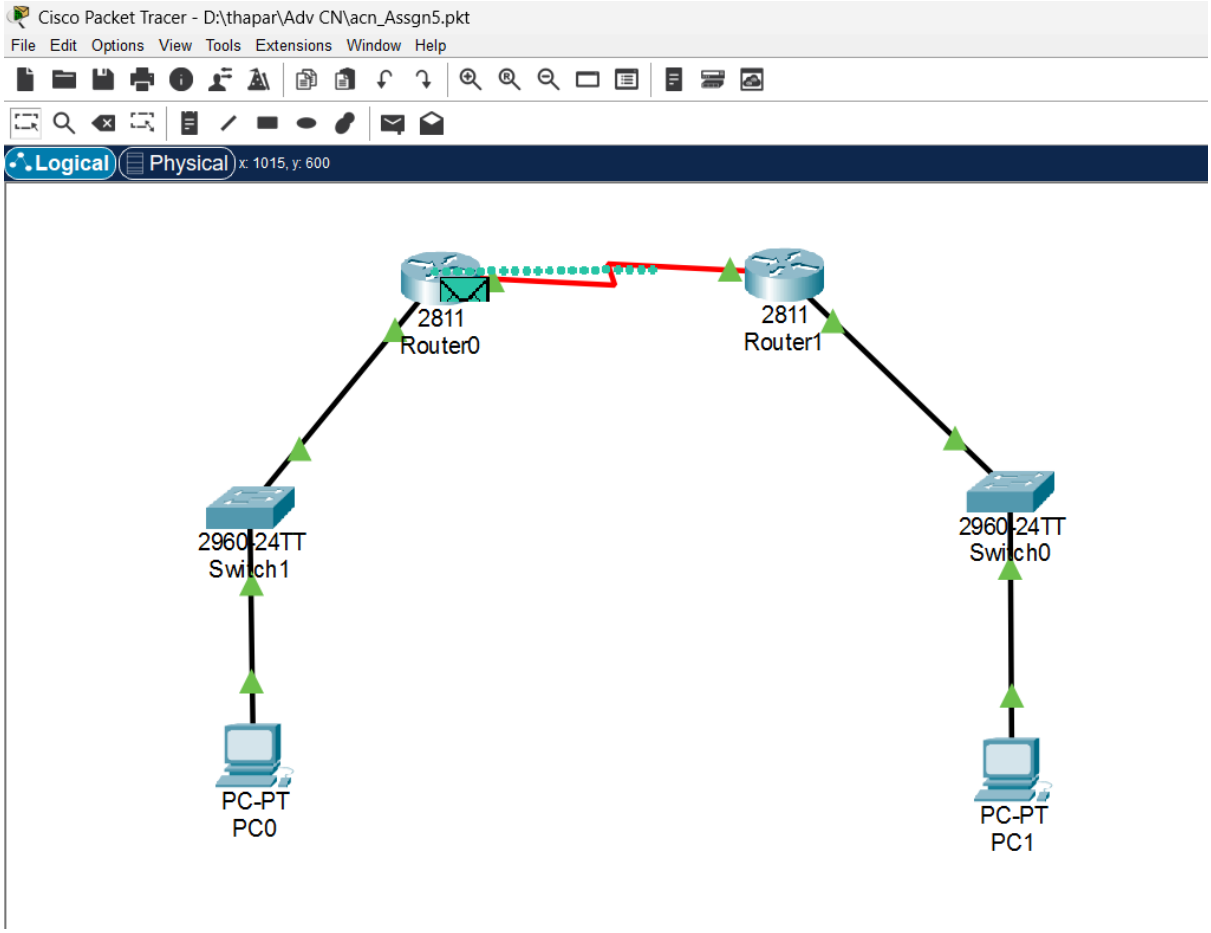


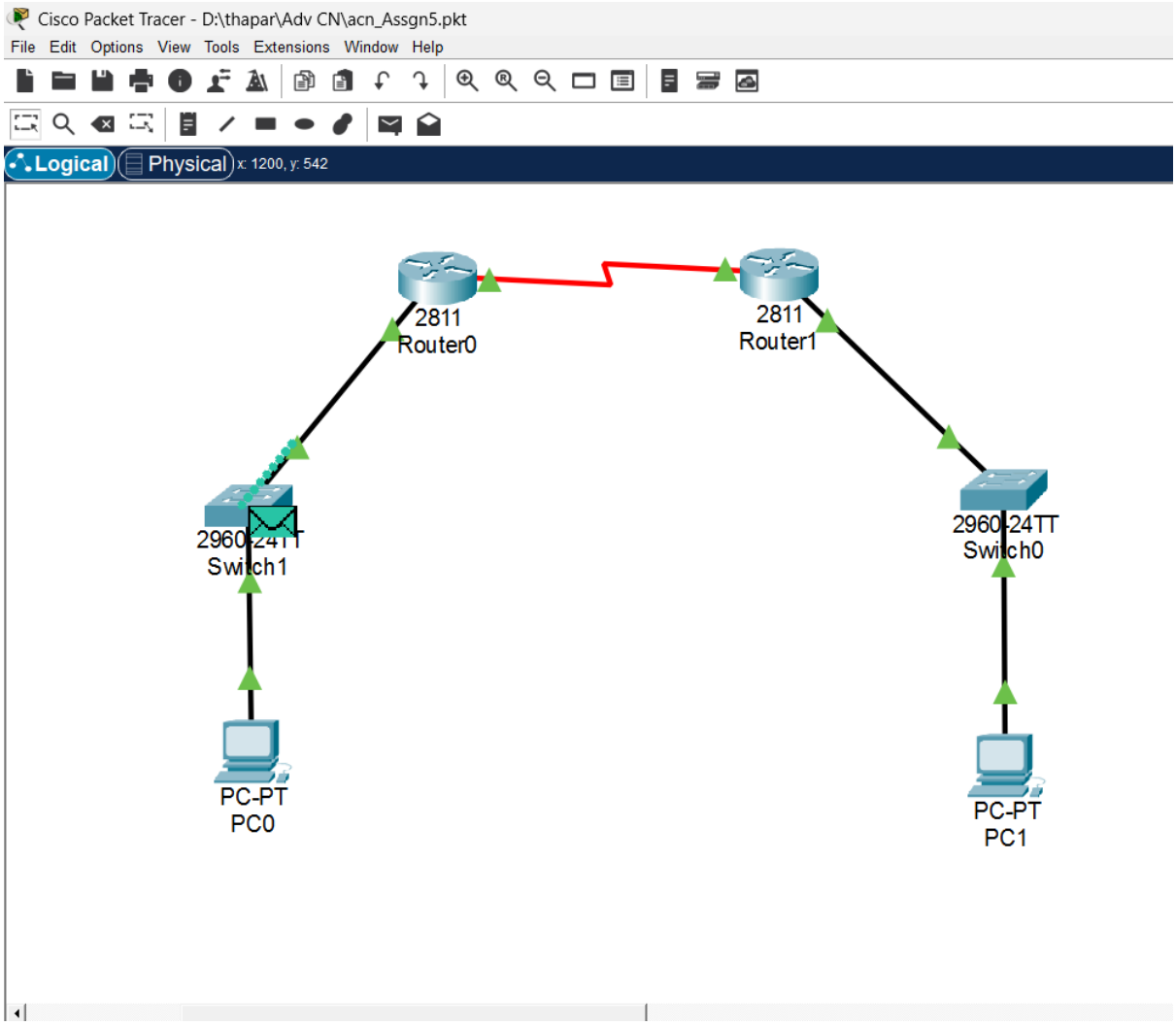
Pinging PC0 (192.168.1.100) from PC1 (172.16.0.100)

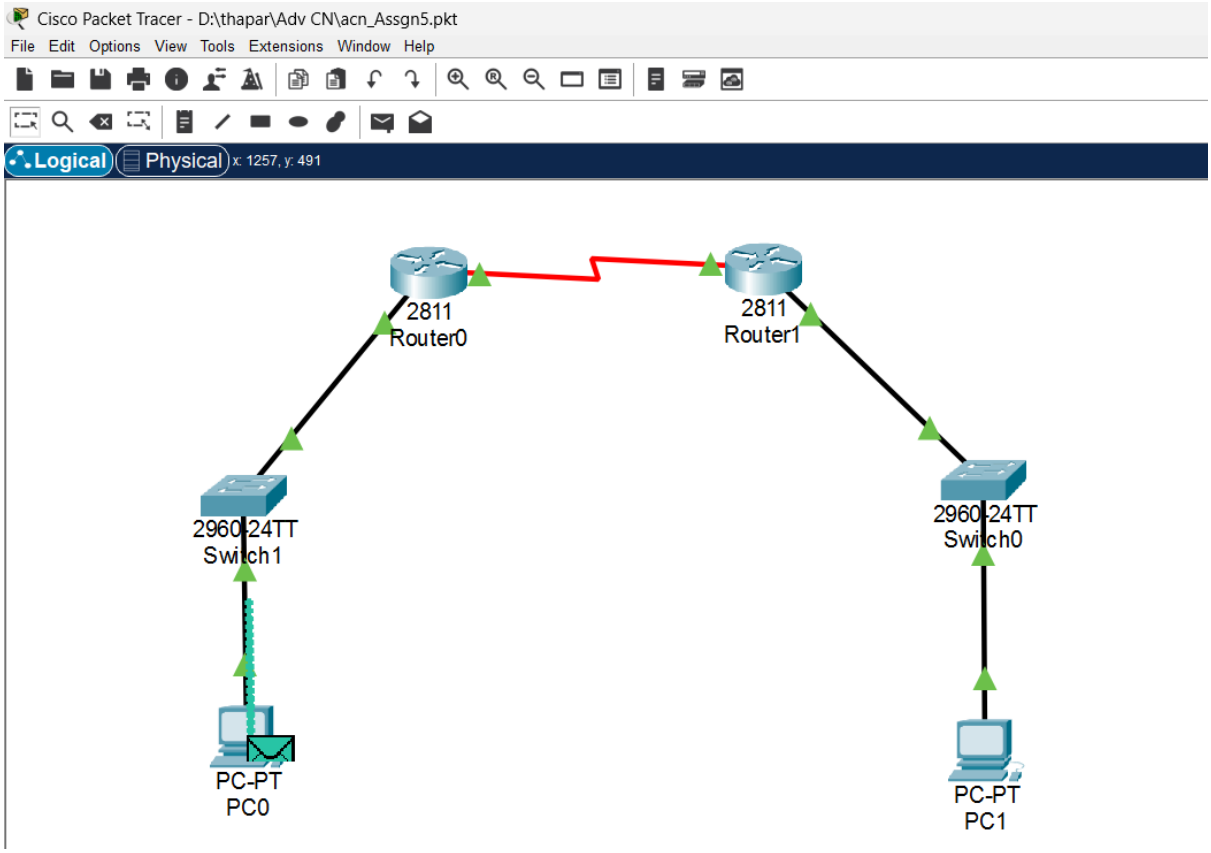

Cisco Packet Tracer - D:\thapar\Adv CN\acn_Assgn5.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1166, y: 417
2811
Router0
2811
Router1
2960-24TT
Switch1
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1

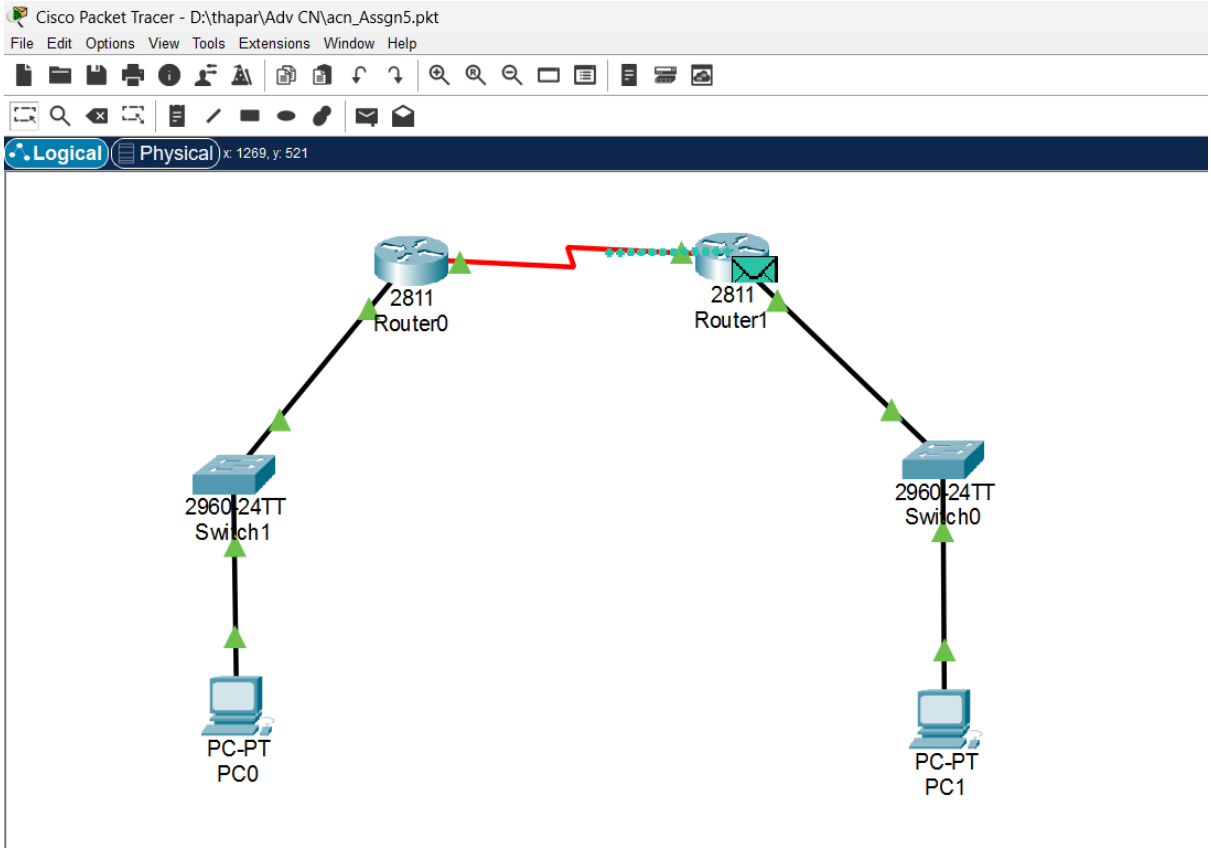

Cisco Packet Tracer - D:\thapar\Adv CN\acn_Assgn5.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1028, y: 488
2811
Router0
2811
Router1
2960-24TT
Switch1
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1

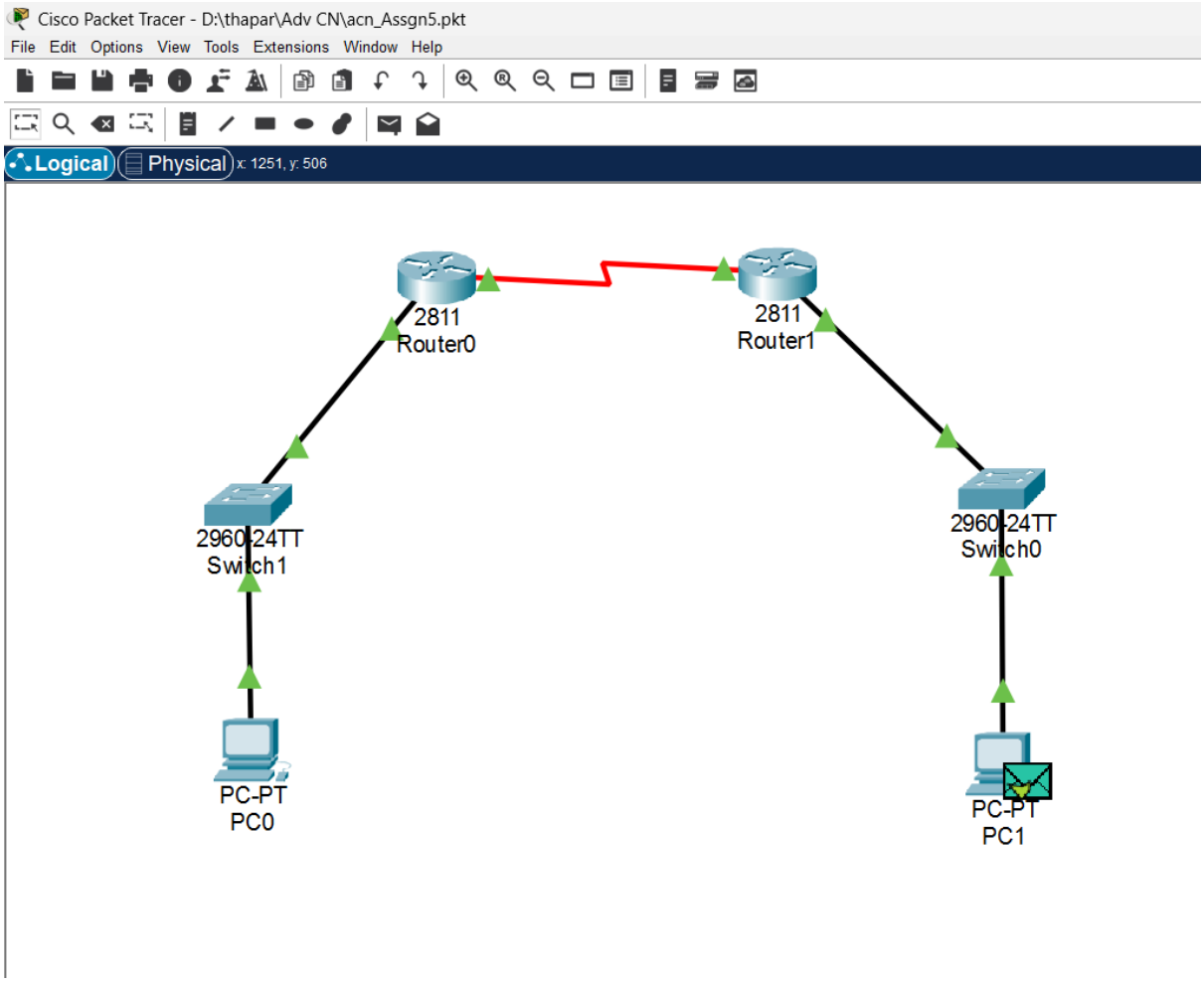

Cisco Packet Tracer - D:\thapar\Adv CN\acn_Assgn5.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1031, y: 491
2811
Router0
2811
Router1
2960-24TT
Switch1
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1

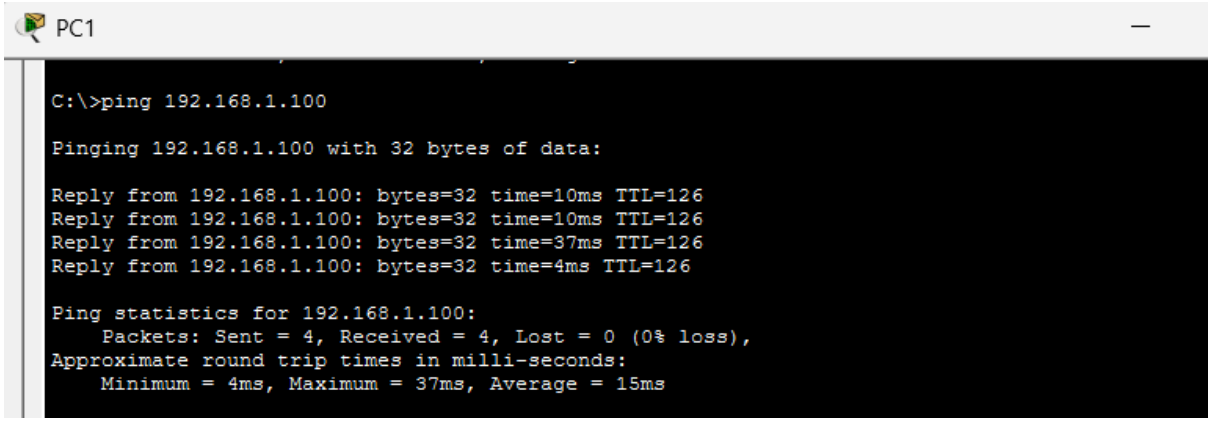

Cisco Packet Tracer - D:\thapar\Adv CN\acn_Assgn5.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1015, y: 600
2811
Router0
2811
Router1
2960-24TT
Switch1
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1


Cisco Packet Tracer - D:\thapar\Adv CN\acn_Assgn5.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1200, y: 542
2811
Router0
2811
Router1
2960-24TT
Switch1
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1


Cisco Packet Tracer - D:\thapar\Adv CN\acn_Assgn5.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1257, y: 491
2811
Router0
2811
Router1
2960-24TT
Switch1
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1


Cisco Packet Tracer - D:\thapar\Adv CN\acn_Assgn5.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1269, y: 521
2811
Router0
2811
Router1
2960-24TT
Switch1
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1


Cisco Packet Tracer - D:\thapar\Adv CN\acn_Assgn5.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1251, y: 506
2811
Router0
2811
Router1
2960-24TT
Switch1
2960-24TT
Switch0
PC-PT
PC0
PC-PT
PC1


PC1
C:\>ping 192.168.1.100
Pinging 192.168.1.100 with 32 bytes of data:
Reply from 192.168.1.100: bytes=32 time=10ms TTL=126
Reply from 192.168.1.100: bytes=32 time=10ms TTL=126
Reply from 192.168.1.100: bytes=32 time=37ms TTL=126
Reply from 192.168.1.100: bytes=32 time=4ms TTL=126
Ping statistics for 192.168.1.100:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 37ms, Average = 15ms